

KUAN BUTTS

geomatics, spatial computation, network science, OSS, data science/engineering

EXPERIENCE

Mapbox – Staff Software Engineer

NOV 2019 – PRESENT, SAN FRANCISCO, CA

SENIOR SWE: NOV 2019 – APR 2025

- Processing and classification of live global stream of multimodal location data
- Calculation of real-time road speed profiles, feeding ETA and route selection
- Developed distributed inference pipeline for real-time global road segment speed estimates, handling >5B observations/hour
- Migrated live speeds to decoupled, high-accuracy probe-matching system, improving ETA precision by +5.9% and recall by +10.8%. Optimized modality classifier, achieved speed accuracy F1 gains of 2–11% (region dependent) via revamped matching system; yielded up to ~60% increased coverage on surface roads
- Streamlined pipelines for modality classification model training on raw probe data. Developed ML applications on vehicle telemetry to flag routing anomalies.
- Delivered a global, daily-updated activity indices product at road segment and block-level resolution, enabling high-granularity mobility insights.

UrbanFootprint – Data Science & Engineering

JAN 2017 – SEP 2019, BERKELEY, CA

- Optimization of high resolution spatial models with prolonged runtimes
- Development of network graph models on urban travel network data

Code for America – Software Engineer, Fellow

JAN 2016 – DEC 2016, SAN FRANCISCO, CA

- Text-based communications software development (ClientComm)
- Generated large text communications dataset for client success rate research
- Application deployed in 3+ US jurisdictions, serving thousands of end users

Microsoft – Civic Technology Fellow

AUG 2015 – DEC 2015, NEW YORK CITY, NY

- Ingestion of live NYC MTA bus data to monitor real-time schedule adherence

MIT + MIT Media Lab – Graduate Researcher, Software Developer

AUG 2012 – AUG 2015, CAMBRIDGE, MA

- Contributed to development of app to geotag rider sentiment along transit lines

EDUCATION

MIT – Master of City Planning, Transportation Systems Planning

AUG 2012 – SEP 2014, CAMBRIDGE, MA

Thesis: *Iterative Methods in Adapting Mobile Technologies for Data Acquisition, A Case Study in St. Louis, Missouri*

Washington University – BS Architecture, Urban Planning Focus

AUG 2007 - MAY 2011, ST. LOUIS, MO